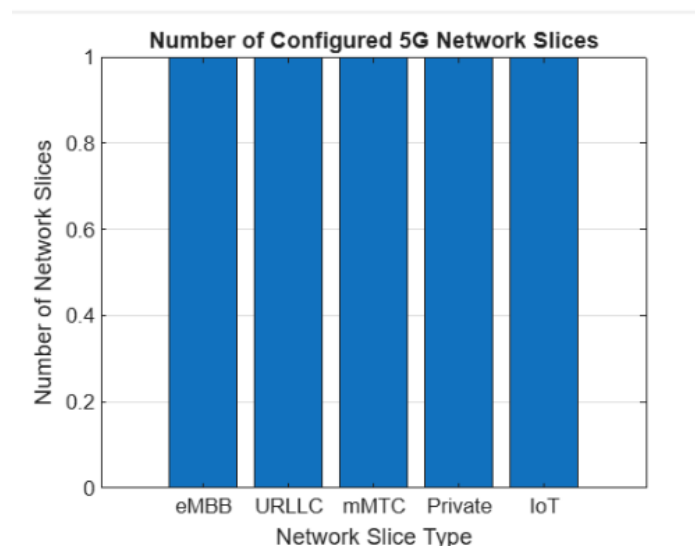


Experiment: 10

5G Network Slicing Environment Using MATLAB

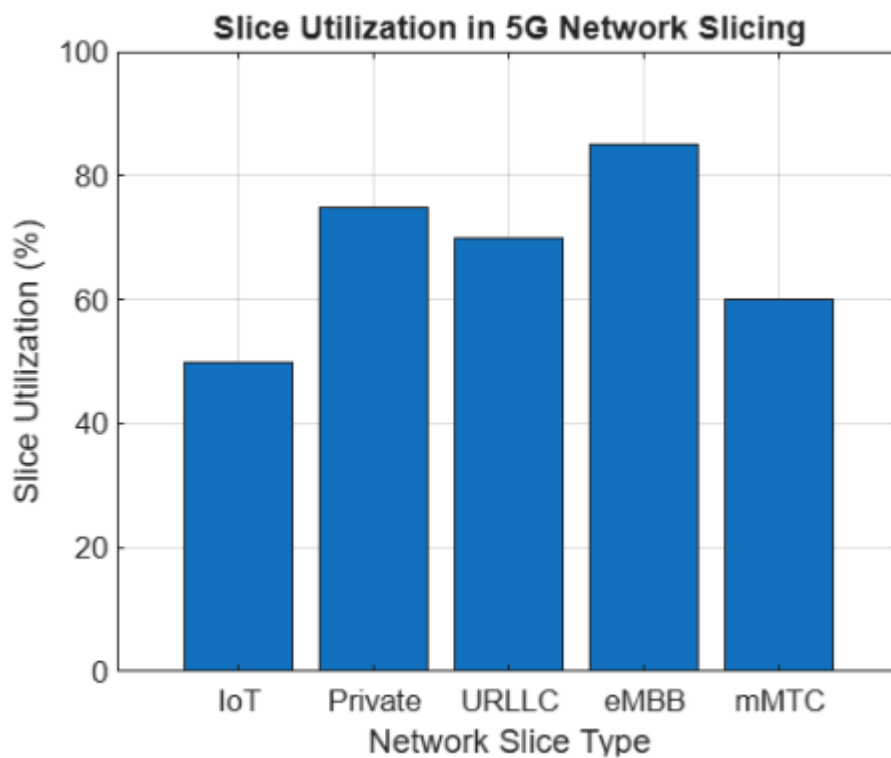
Objective – 1

```
clc;  
clear;  
close all;  
  
slices = {'eMBB','URLLC','mMTC','Private','IoT'};  
count = [1 1 1 1 1]; % One instance of each network slice  
  
figure  
bar(count)  
grid on  
set(gca,'XTickLabel',slices)  
xlabel('Network Slice Type')  
ylabel('Number of Network Slices')  
title('Number of Configured 5G Network Slices')
```



Objective – 2

```
clc;  
clear;  
close all;  
  
slices = categorical({'eMBB','URLLC','mMTC','Private','IoT'});  
utilization = [85 70 60 75 50];  
  
figure;  
bar(slices, utilization);  
grid on;  
xlabel('Network Slice Type');  
ylabel('Slice Utilization (%)');  
title('Slice Utilization in 5G Network Slicing');  
ylim([0 100]);
```



Objective – 3

```
clc;  
clear;  
close all;  
slices = [1 2 3 4 5];  
latency = [25 5 40 15 60];  
figure;  
plot(slices, latency, '-o','LineWidth',2,'MarkerSize',8);  
grid on;  
xticks(slices);  
xticklabels({'eMBB','URLLC','mMTC','Private','IoT'});  
xlabel('Network Slice Type');  
ylabel('End-to-End Latency (ms)');  
title('5G Network Slicing: End-to-End Latency');  
ylim([0 70]);
```

